

hovid **ERECOR ORODISPERSIBLE FILMS**
Sildenafil Citrate Orodispersible Films 50mg

VIERExx-06 (MY)

DESCRIPTION

Blue colored, rectangular, opaque, non sticky, orally disintegrating strip.

COMPOSITION

Each orodispersible film contains Sildenafil Citrate Ph. Eur. equivalent to Sildenafil 50mg.

PHARMACODYNAMICS

Sildenafil is a selective inhibitor of cyclic guanosine monophosphate (cGMP) -specific phosphodiesterase type 5 (PDE5).

The physiologic mechanism of erection of the penis involves release of nitric oxide (NO) in the corpus cavernosum during sexual stimulation.

NO then activates the enzyme guanylate cyclase, which results in increased levels of cyclic guanosine monophosphate (cGMP), producing smooth muscle relaxation in the corpus cavernosum and allowing inflow of blood.

Sildenafil has no direct relaxant effect on isolated human corpus cavernosum, but enhances the effect of NO by inhibiting of phosphodiesterase type 5 (PDE5), which is responsible for degradation of cGMP in the corpus cavernosum.

When sexual stimulation causes local release of NO, inhibition of PDE5 by sildenafil causes increased levels of cGMP in the corpus cavernosum, resulting in smooth muscle relaxation and inflow of blood to the corpus cavernosum.

Sildenafil at recommended doses has no effect in the absence of sexual stimulation.

Sildenafil is selective for PDE5. Its effect is more potent on PDE5 than on other known phosphodiesterases (10-fold for PDE6, >80-fold for PDE1, >700-fold for PDE2, PDE3, and PDE4, PDE7 - PDE11).

The approximately 4,000-fold selectivity for PDE5 versus PDE3 is important because PDE3 is involved in the control of cardiac contractility.

PHARMACOKINETICS

Sildenafil pharmacokinetics are dose-proportional over the recommended dose range. It is eliminated predominantly by hepatic metabolism (mainly cytochrome P450 3A4) and is converted to an active metabolite with properties similar to the parent, sildenafil.

Absorption

Sildenafil is rapidly absorbed after oral administration, with mean absolute bioavailability of 41% (range 25%-63%).

Sildenafil inhibits the human PDE5 enzyme *in vitro* by 50% at a concentration of 3.5 nM. In man, the mean maximum free plasma concentration of sildenafil following a single oral dose of 100 mg is approximately 18 ng/mL, or 38 nM.

Maximum observed plasma concentrations are reached within 30 to 120 minutes (median 60 minutes) of oral dosing in the fasted state.

When sildenafil film-coated tablets are taken with a high fat meal, the rate of absorption of sildenafil is reduced, with a mean delay in Tmax of 60 minutes and a mean reduction in Cmax of 29%, however, the extent of absorption was not significantly affected (AUC decreased by 11%).

Distribution

The mean steady-state volume of distribution (V_{ss}) for sildenafil is 105 L, indicating distribution into the tissues.

Sildenafil and its major circulating N-desmethyl metabolite are both approximately 96% bound to plasma proteins. Protein binding is independent of total drug concentrations.

Based upon measurements of sildenafil in semen of healthy volunteers 90 minutes after dosing, less than 0.0002% (average 188 ng) of the administered dose may appear in the semen of patients.

Metabolism

Sildenafil is cleared predominantly by the CYP3A4 (major route) and CYP2C9 (minor route) hepatic microsomal isoenzymes. The major circulating metabolite results from N-desmethylation of sildenafil, and is itself further metabolized. This metabolite has a PDE selectivity profile similar to sildenafil and an *in vitro* potency for PDE5 approximately 50% of the parent drug. Plasma concentrations of this metabolite are approximately 40% of those seen for sildenafil in healthy person. The N-desmethyl metabolite is further metabolized, with a terminal half-life of approximately 4 hours.

Elimination

The total body clearance of sildenafil is 41 L/h with a resultant terminal phase half-life of 3-5 hours. After either oral or intravenous administration, sildenafil is excreted as metabolites predominantly in the feces (approximately 80% of administered oral dose) and to a lesser extent in the urine (approximately 13% of the administered oral dose).

Special Population

Elderly

Healthy elderly volunteers (65 years or over) had a reduced clearance of sildenafil, resulting in approximately 90% higher plasma concentrations of sildenafil and the active N-desmethyl metabolite compared to healthy younger person (18-45 years). Due to age-differences in plasma protein binding, the corresponding increase in free sildenafil plasma concentration was approximately 40%.

Renal Insufficiency

In patients with mild (creatinine clearance = 50-80 mL/min) and moderate (creatinine clearance = 30-49 mL/min) renal impairment, the pharmacokinetics of a single oral dose of sildenafil (50 mg) were not altered.

In patients with severe (creatinine clearance = <30 mL/min) renal impairment, sildenafil clearance was reduced, resulting in approximately doubling of AUC (100%) and C_{max} (88%). In addition, N-desmethyl metabolite AUC and C_{max} values were significantly increased by 200% and 79% respectively in patients with severe renal impairment.

Hepatic Insufficiency

In patients with hepatic cirrhosis (Child-Pugh A and B), sildenafil clearance was reduced, resulting in increases in AUC (85%) and C_{max} (47%). The pharmacokinetics of sildenafil in patients with severely impaired hepatic function (Child-Pugh class C) have not been studied.

INDICATIONS

Sildenafil is indicated for the treatment of erectile dysfunction, which is the inability to achieve or maintain a penile erection sufficient for satisfactory sexual performance.

In order for sildenafil to be effective, sexual stimulation is required.

CONTRAINDICATIONS

Use of sildenafil is contraindicated in patients with a known hypersensitivity to any component of the orodispersible films.

Sildenafil was shown to potentiate the hypotensive effects of acute and chronic nitrates, and its administration to patients who are concurrently using nitric oxide donors, organic nitrates or organic nitrites in any form either regularly or intermittently is therefore contraindicated.

The co-administration of PDE5 inhibitors, including sildenafil, with guanylate cyclase stimulators, such as riociguat, is contraindicated as it may potentially lead to symptomatic hypotension.

WARNING AND PRECAUTIONS

A thorough medical history and physical examination should be undertaken to diagnose erectile dysfunction, determine potential underlying causes, and identify appropriate treatment.

There is a degree of cardiac risk associated with sexual activity; therefore, physicians may wish to consider the cardiovascular status of their patients prior to initiating any treatment for erectile dysfunction.

Agents for the treatment of erectile dysfunction should not be used in men for whom sexual activity is inadvisable.

Serious cardiovascular events, including myocardial infarction, sudden cardiac death, ventricular arrhythmia, cerebrovascular hemorrhage and transient ischemic attack are in temporal association with the use of sildenafil for erectile dysfunction. Most, but not all, of these patients had pre-existing cardiovascular risk factors. Many of these events were reported to occur during or shortly after sexual activity, a few were reported to occur shortly after the use of sildenafil without sexual activity, and others were reported to have occurred hours to days after the use of sildenafil and sexual activity. It is not possible to determine whether these events are related directly to sildenafil, to sexual activity, to the patient's underlying cardiovascular disease, to a combination of these factors, or to other factors.

Sildenafil has been shown to have systemic vasodilatory properties that result in transient decreases in blood pressure. This is of little or no consequence in most patients. However, prior to prescribing sildenafil, physicians should carefully consider whether their patients with certain underlying conditions could be adversely affected by such vasodilatory effects, especially in combination with sexual activity. Patients with increased susceptibility to vasodilators include those with left ventricular outflow obstruction (e.g., aortic stenosis, hypertrophic obstructive cardiomyopathy), or those with the rare syndrome of multiple system atrophy manifesting as severely impaired autonomic control of blood pressure.

Non-arteritic anterior ischemic optic neuropathy (NAION), a rare condition and a cause of decreased vision or loss of vision, has been reported rarely post-marketing with the use of all PDE5 inhibitors, including sildenafil. In case of sudden visual loss, patients should be advised to stop taking sildenafil and consult a physician immediately. Individuals who have already experienced NAION are at increased risk of NAION recurrence. Therefore physicians should discuss this risk with these patients and whether they could be adversely affected by use of PDE5 inhibitors. PDE5 inhibitors, including sildenafil, should be used with caution in these patients and only when the anticipated benefits outweigh the risks.

Caution is advised when sildenafil is administered to patients taking an alpha-blocker, as the co-administration may lead to symptomatic hypotension in a few susceptible individuals. In order to minimize the potential for developing postural hypotension, patients should be hemodynamically stable on alpha-blocker therapy prior to initiating sildenafil treatment. Initiation of sildenafil at lower doses should be considered. In addition, physicians should advise patients what to do in the event of postural hypotensive symptoms.

A minority of patients with the inherited condition retinitis pigmentosa have genetic disorders of retinal phosphodiesterases. There is no safety information on the administration of sildenafil to patients with retinitis pigmentosa, therefore, sildenafil should be administered with caution to these patients.

Sildenafil may potentiate the anti-aggregatory effect of sodium nitroprusside (a nitric oxide donor). There is no safety information on the administration of sildenafil to patients with bleeding disorders or active peptic ulceration, therefore sildenafil should be administered with caution to these patients.

Agents for the treatment of erectile dysfunction should be used with caution in patients with anatomical deformation of the penis (such as angulation, cavernosal fibrosis or Peyronie's disease), or in patients who have conditions which may predispose them to priapism (such as sickle cell anemia, multiple myeloma or leukemia).

Prolonged erections and priapism may occur with sildenafil. In the event of an erection that persists longer than 4 hours, the patient should seek immediate medical assistance. If priapism is not treated immediately, penile tissue damage and permanent loss of potency could result.

The safety and efficacy of combinations of sildenafil with other PDE5 inhibitors, or other pulmonary arterial hypertension (PAH) treatments containing sildenafil, or other treatments for erectile dysfunction is not recommended.

Sudden decrease or loss of hearing may occur with the use of all PDE5 inhibitors, including sildenafil. No causal relationship has been made between the use of PDE5 inhibitors and sudden decrease or loss of hearing. In case of sudden decrease or loss of hearing patients should be advised to stop taking sildenafil and consult a physician promptly.

PREGNANCY AND LACTATION

Sildenafil is not indicated for use by women. There is no adequate and well controlled studies in pregnant and breastfeeding women.

DRUG INTERACTIONS

Effects of other medicinal products on sildenafil
Sildenafil metabolism is principally mediated by the cytochrome P450 (CYP) isoforms 3A4 (major route) and 2C9 (minor route). Therefore, inhibitors of these isoenzymes may reduce sildenafil clearance and inducers of these isoenzymes may increase sildenafil clearance.

Cimetidine, a cytochrome P450 inhibitor and a non-specific CYP3A4 inhibitor, caused increase in plasma sildenafil concentrations when co-administered with sildenafil. When sildenafil was administered with erythromycin, a moderate CYP3A4 inhibitor, there was an increase in sildenafil systemic exposure (AUC). In addition, co-administration of the HIV protease inhibitor saquinavir, also a CYP3A4 inhibitor, resulted in an increase in sildenafil Cmax and AUC. Sildenafil had no effect on saquinavir pharmacokinetics. Stronger CYP3A4 inhibitors such as ketoconazole and itraconazole would be expected to have greater effects.

Co-administration with the HIV protease inhibitor ritonavir, which is a highly potent P450 inhibitor, with sildenafil resulted in increase in sildenafil Cmax and plasma AUC. Sildenafil had no effect on ritonavir pharmacokinetics. It is generally well tolerated when the dose of sildenafil for patients receiving potent CYP3A4 inhibitors was administered as recommended.

Single doses of antacid (magnesium hydroxide/aluminium hydroxide) did not affect the bioavailability of sildenafil.

Co-administration of the endothelin antagonist, bosentan, (an inducer of CYP3A4 [moderate], CYP2C9 and possibly of CYP2C19 with sildenafil resulted in decrease in sildenafil AUC and Cmax, while sildenafil increased bosentan AUC and Cmax. Concomitant administration of strong CYP3A4 inducers, such as rifampin, is expected to cause greater decreases in plasma concentrations of sildenafil.

CYP2C9 inhibitors (such as tolbutamide, warfarin), CYP2D6 inhibitors (such as selective serotonin reuptake inhibitors, tricyclic antidepressants), thiazide and related diuretics, angiotensin converting enzyme (ACE) inhibitors, and calcium channel blockers has shown no effects on sildenafil pharmacokinetics.

Azithromycin does not show effect on the AUC, Cmax, Tmax, elimination rate constant, or subsequent half-life of sildenafil or its major circulating metabolite.

Effects of sildenafil on other medicinal products.
Sildenafil is a weak inhibitor of the cytochrome P450 isoforms 1A2, 2C9, 2C19, 2D6, 2E1 and 3A4 (IC50 > 150 µ M). It is unlikely that sildenafil will alter the clearance of substrates of these isoenzymes.
Sildenafil may potentiate the hypotensive effect of acute and chronic nitrates. Therefore, use of nitric oxide donors, organic nitrates, or organic nitrites in any form either regularly or intermittently with sildenafil is contraindicated.

When sildenafil and doxazosin were administered simultaneously to patients stabilized on doxazosin therapy, patients may experience symptomatic postural hypotension, including dizziness and light-headedness, but not syncope. Concomitant administration of sildenafil to patients taking alpha-blocker therapy may lead to symptomatic hypotension in a few susceptible Individuals.

There is no significant interactions when sildenafil is co-administered with tolbutamide or warfarin, both of which are metabolized by CYP2C9.

Sildenafil do not affect the steady-state pharmacokinetics of the HIV protease inhibitors, saquinavir and ritonavir, both of which are CYP3A4 substrates.

Sildenafil resulted in increase in bosentan AUC and Cmax. Sildenafil do not potentiate the increase in bleeding time caused by aspirin. Sildenafil do not potentiate the hypotensive effect of alcohol in healthy person.

There is no interaction when sildenafil was co-administered with amlodipine in hypertensive patients.

MAIN SIDE/ADVERSE EFFECTS

The adverse events were generally transient and mild to moderate in nature.

Infections and infestations

Uncommon : Rhinitis

Immune system disorders

Uncommon : Hypersensitivity

Nervous system disorders

Very common : Headache
Common : Dizziness
Uncommon : Somnolence
Rare : Seizure, Seizure recurrence, Syncope

Eye disorders

Common : Vision blurred, Visual disturbance, Cyanopsia
Uncommon : Eye pain, Photophobia, Photopsia, Chromatopsia, Ocular hyperaemia, Visual brightness
Rare : Eye oedema, Eye swelling, Dry eye, Asthenopia, Halo vision, Xanthopsia, Erythropsia, Eye disorder, Conjunctival hyperaemia, Eye irritation, Abnormal sensation in eye, Eyelid oedema

Cardiac disorders

Uncommon : Tachycardia, Palpitations

Vascular disorders

Common : Hot flush, Flushing
Uncommon : Hypotension

Respiratory, thoracic and mediastinal disorders

Common : Nasal congestion
Uncommon : Epistaxis, Sinus congestion
Rare : Throat tightness, Nasal dryness, Nasal oedema

Gastrointestinal disorders

Common : Nausea, Dyspepsia
Uncommon : Gastro oesophageal reflux disease, Vomiting, Abdominal pain upper, Dry mouth
Rare : Hypoaesthesia oral

Skin and subcutaneous tissue disorders

Uncommon : Rash

Musculoskeletal and connective tissue disorders

Uncommon : Myalgia, Pain in extremity

Reproductive system and breast disorders

Rare : Priapism, Erection increased

General disorder and administration site conditions

Uncommon : Feeling hot
Rare : Irritability

Investigations

Uncommon : Heart rate increased

OVERDOSE AND TREATMENT

In cases of overdose, standard supportive measures should be adopted as required. Renal dialysis is not expected to accelerate clearance as sildenafil is highly bound to plasma proteins and it is not eliminated in the urine.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

No studies on the effects on the ability to drive and use machines have been performed.

As dizziness and altered vision may occur with Sildenafil, patients should be aware of how they react to Sildenafil before driving or operating machinery.

DOSAGE AND ADMINISTRATION

Adults

The recommended dose is 50 mg taken as needed approximately one hour before sexual activity. Based on efficacy and toleration, the dose may be increased to 100 mg or decreased to 25 mg. The maximum recommended dose is 100 mg. The maximum recommended dosing frequency is once per day.

Elderly

Dosage adjustments are not required in elderly patients.

Renal Impairment

Dosing adjustments are not required in patients with mild to moderate renal impairment (creatinine clearance = 30 -80 mL/min).

Since Sildenafil clearance is reduced in patients with severe renal impairment (creatinine clearance <30 mL/min) a 25 mg dose should be considered.

Hepatic Impairment

Since Sildenafil clearance is reduced in patients with hepatic impairment (e.g. cirrhosis), a 25 mg dose should be considered.

Pediatric Population

Sildenafil is not indicated for individuals below 18 years of age.

Use in patients using other medicines

For patients receiving concomitant therapy with ritonavir, it is recommended not to exceed a maximum single dose of 25 mg of sildenafil in a 48-hour period.

A starting dose of 25 mg should be considered in patients receiving concomitant treatment with the CYP3A4 inhibitors (e.g., erythromycin, saquinavir, ketoconazole, itraconazole).

In order to minimize the potential for developing postural hypotension, patients should be stable on alpha-blocker therapy prior to initiating sildenafil treatment. In addition, initiation of sildenafil at lower doses should be considered.

METHOD OF ADMINISTRATION

For oral use only.



Step 1.
Hold the sachet with backside facing you. Locate the arrow on the side, which is at the top left corner of the sachet, and hold it with the side facing up. The sachet is partially sealed on this side with two exposed flaps.



Step 2.
Hold the two exposed flaps of the sachet at the arrow mark. With a gentle peeling motion, start pulling apart in opposite directions both flaps of the sachet at this point and thereby separating the two layers of the sachet.



Step 3.
Continue the peeling motion until the two layers of the sachet are now almost separated. The orodispersible film is now visible inside the sachet. There is no need to completely pull apart both the layers of the sachet to access the orodispersible film.



Step 4, 5, and 6.
Carefully remove the orodispersible film from the sachet and put the orodispersible film on top of your tongue. It will dissolve by itself within seconds.

Important: Do not handle the orodispersible film with wet hands!

Storage : Store below 30°C.
Protect from light and moisture.

Presentation : 4 x 1 Film

Product
Registration Holder : Hovid Bhd.
121, Jalan Tunku Abdul Rahman
(Jalan Kuala Kangsar),
30010 Ipoh, Perak, Malaysia.

Manufactured by : ZIM LABORATORIES LIMITED
B-21/22, MIDC Area, Kalmeshwar,
Nagpur 441 501, Maharashtra State,
India.

Revision date : December 2022